

CA5 Question 1 Report

Legal Retrieval Pipeline

tahamajs

Student ID: 810101504

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1 Summary

This report documents the final implementation of the Q1 legal retrieval system. The pipeline is fully modularized in Python modules under `answers/Q1/src`, executed in strict live mode, and produces reproducible artifacts for indexing, timing, evaluation, and plotting.

2 Implementation Outputs

Artifact	Description
<code>answers/Q1/data/indexed_dataset.pkl</code>	Serialized TF-IDF index and metadata
<code>answers/Q1/data/pipeline_timings.csv</code>	End-to-end timing metrics per benchmark query
<code>answers/Q1/data/rag_evaluation_report.xlsx</code>	RAGAS metrics exported to Excel
<code>answers/reports/figures/q1/*.png</code>	Report-ready plots generated from metrics

3 Pipeline Design

The Q1 system follows this deterministic flow:

1. Dataset loading and OCR-aware text extraction
2. Chunking and metadata assignment
3. TF-IDF index build and persistence
4. Query rewrite, intent classification, metadata extraction
5. Retrieval, reranking, and evidence-grounded response generation
6. Timing capture and benchmark export
7. Live RAGAS evaluation with OpenAI
8. Plot generation for report publication

4 Figures

Missing figure: `figures/q1/avg_node_timings.png`

Figure 1: Average node timing

Missing figure: `figures/q1/node_timing_distributions.png`

Figure 2: Node timing distributions

Missing figure: `figures/q1/total_latency_histogram.png`

Figure 3: Total latency distribution

Missing figure: `figures/q1/rag_eval_metric_means.png`

Figure 4: Mean RAGAS metric scores

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Figure 5: RAGAS score distributions

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Figure 6: Faithfulness vs answer relevancy

5 Reproducible Execution

All commands are executed under:

```
source /Users/tahamajs/Documents/uni/venv/bin/activate
```

Primary command:

```
bash answers/scripts/run_all.sh
```

6 Validation Criteria

The final implementation targets the following checks:

- Strict live preflight passes with all required keys and tools.
- Notebook execution completes from top to bottom.
- Required index, timing, and evaluation artifacts are non-empty.
- All Q1 figures are generated under `answers/reports/figures/q1`.